

From a Year Behind to Near-Real-Time: Watching the Ocean Carbon Sink as It Happens

Carbon Monitor Ocean (CMO-NRT) uses CNNs and semi-supervised learning to update ten ocean biogeochemical models and eight data products to the present month. From Tsinghua University, Microsoft Research, the University of Exeter and collaborators.

The ocean is one of the planet's largest buffers against climate change, soaking up roughly a quarter of the carbon dioxide humanity emits every year. Yet our best official accounting of how much it absorbs, the annual Global Carbon Budget, always trails reality by about a year. When the ocean's uptake shifts, we cannot see it until long after the fact.

That lag is not an oversight; it is built into how the number is produced. The ocean biogeochemical models (GOBMs) that estimate the sink are computationally heavy, and the surface-ocean CO₂ observations that data-driven products depend on are themselves delayed by roughly a year. So the community has detailed, trustworthy databases of ocean carbon fluxes that are always looking backward in time.

This is an awkward gap at exactly the wrong moment. As countries move through the global stocktake under the Paris Agreement, they need to know where the carbon is going now, not a year ago. The paper's contribution is a way to close that gap: Carbon Monitor Ocean, or CMO-NRT, a near-real-time, monthly, gridded dataset of global surface-ocean CO₂ fugacity (fCO₂) and air-sea CO₂ flux, together with the deep-learning framework that produces it.

Links

Paper: <https://arxiv.org/abs/2312.01637>

Code: <https://github.com/kepiyu/CMO-NRT>

Dataset (Figshare): <https://doi.org/10.6084/m9.figshare.24658494.v1> | Live project: <https://carbonsink.microsoft.com/>

The key design choice is what CMO-NRT does not do: it does not try to replace the trusted community models. Instead it learns to reproduce them from inputs that are available in near-real-time. Each GOBM or data product becomes a target that a neural network learns to predict from observable quantities, so that once those observables arrive for the current month, the model can extend the estimate forward without waiting for the original, slow pipeline to catch up.

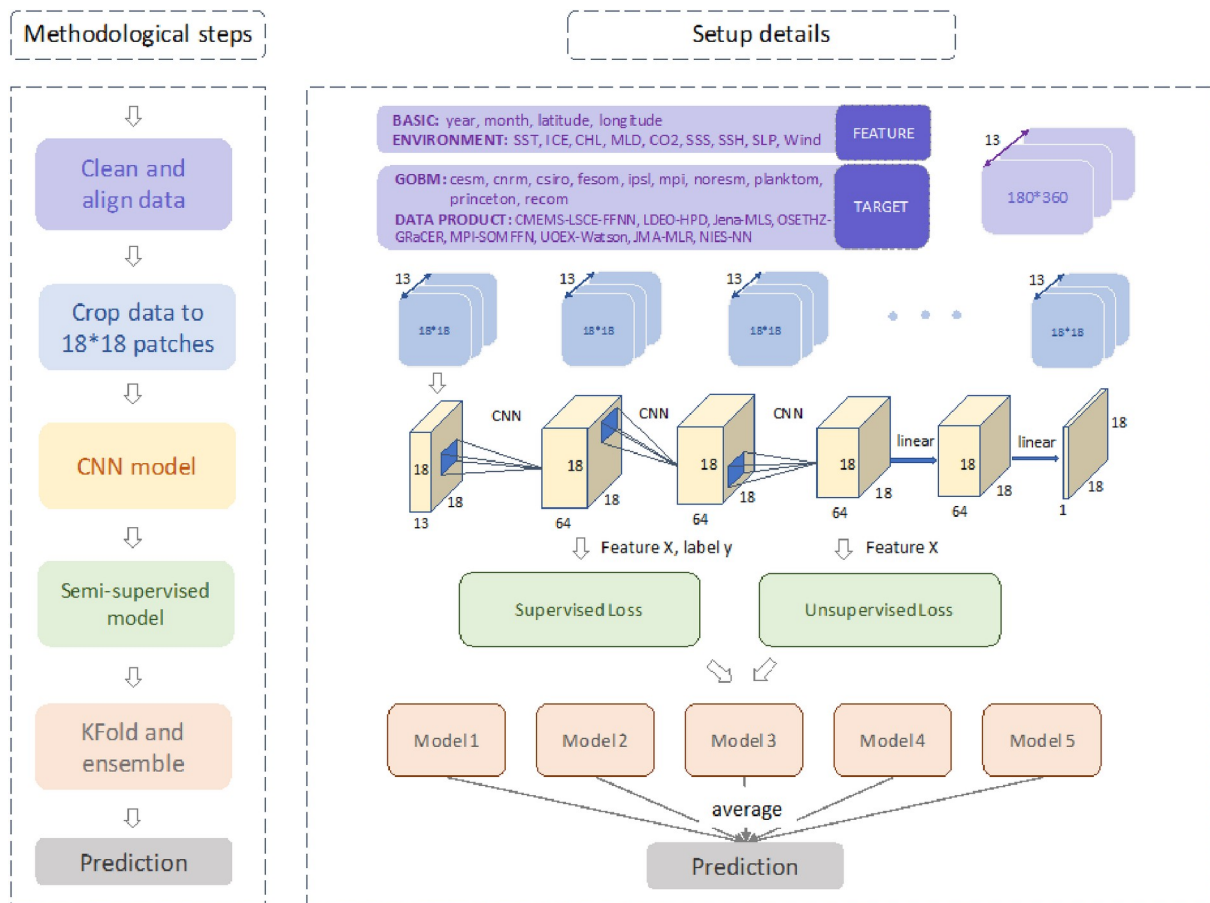


Figure 1. End-to-end overview of CMO-NRT. Left: the five methodological steps, from cleaning and aligning data, cutting it into patches, and training a CNN with semi-supervised learning, to K-fold ensembling and prediction. Right: the setup details, showing the observable features (year, month, latitude, longitude, and nine environmental variables) that the model uses to reproduce each of the ten GOBMs and eight data products on the global 180x360 grid.

The method: learn the models, then run ahead of them

The predictors are deliberately mundane and, crucially, timely: year, month, latitude, longitude, and nine environmental variables such as sea-surface temperature (SST), sea ice (ICE), chlorophyll (CHL), mixed-layer depth (MLD), atmospheric CO₂, salinity (SSS), sea-surface height (SSH), sea-level pressure (SLP), and wind. The global 180x360 grid is cut into small 18x18 patches and fed through a stack of convolutional layers followed by linear layers, which map the nine input channels through hidden widths of 64 down to a single predicted value per grid cell.

The interesting part is how the model is trained. Labeled grid cells give a standard supervised error, but observations are sparse, so the authors add a semi-supervised term. The same patch is shown to the network twice, once weakly augmented with 10% of its features masked and once strongly augmented with 30% masked, and an unsupervised consistency loss penalizes any disagreement between the two predictions. Combining a supervised RMSE loss

with this consistency loss (the total is $L = w \cdot L_u + L_s$) makes the model noticeably more stable where labels run out.

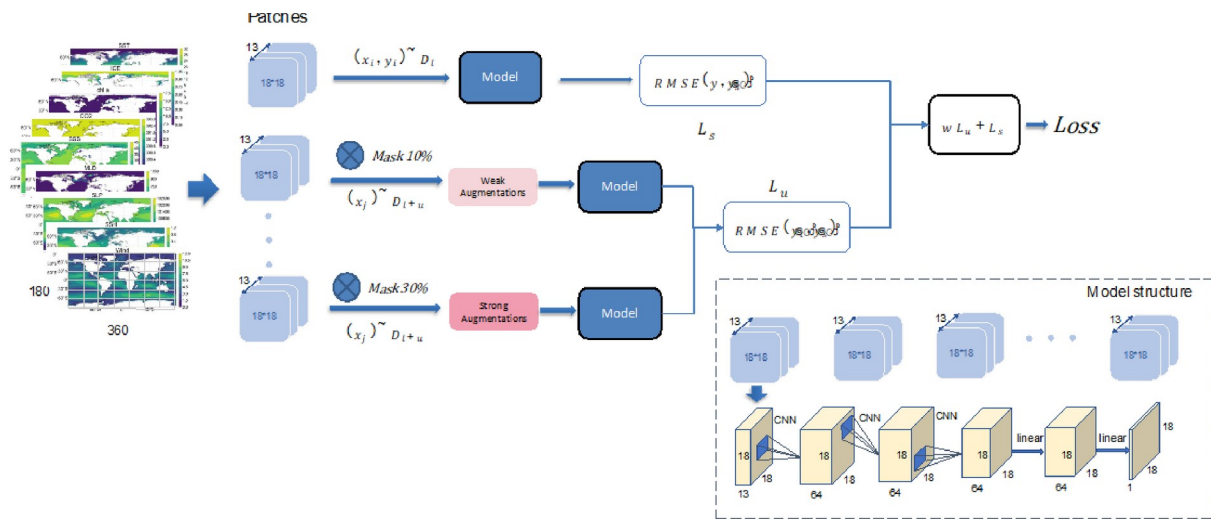


Figure 2. The semi-supervised training scheme. Labeled patches feed a supervised RMSE loss (L_s), while the same unlabeled patches are passed through weak (10% masked) and strong (30% masked) augmentations, and the disagreement between the two predictions forms the unsupervised consistency loss (L_u). The two are combined as $L = w \cdot L_u + L_s$. Inset: the CNN-plus-linear model structure, stacking 18x18 patches through convolutional and linear layers.

Two more engineering details make the system usable in practice. Predictions are ensembled through K-fold cross-validation to reduce variance, and because one key input, atmospheric CO₂ (xCO₂), is itself only available up to the end of 2022, the authors train a separate small model to extend xCO₂ to the present. That auxiliary model reaches a test RMSE of 1.74, about a 0.5% error, which is accurate enough to feed the main pipeline without introducing meaningful drift.

What it produces: the sink, extended to the present month

The payoff is a dataset that carries the ocean carbon record past where the Global Carbon Budget stops. Applying the framework to all ten GOBMs and eight data products of the Global Carbon Budget 2022 yields monthly, gridded maps of surface-ocean fCO₂ and air-sea CO₂ flux running from January 2022 through July 2023, seamlessly continuing the historical series.

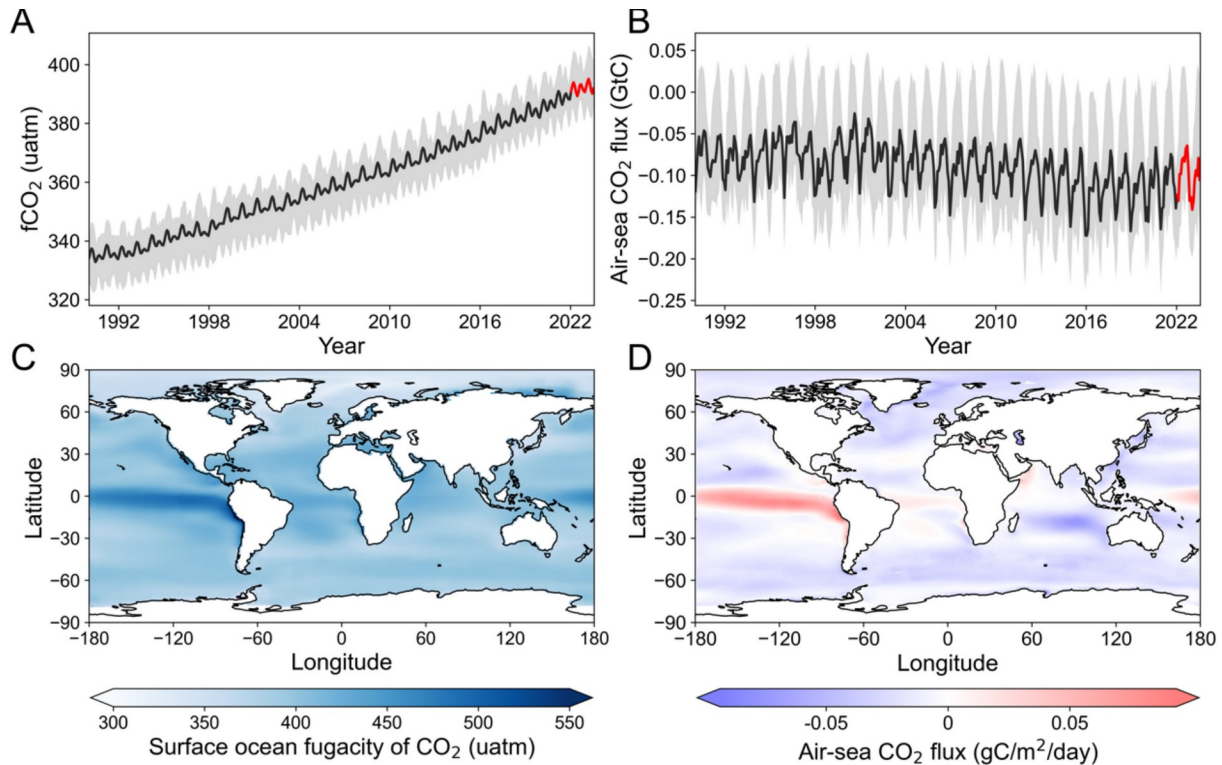


Figure 3. CMO-NRT extends the record to near-real-time. In the $f\text{CO}_2$ (A) and air-sea CO_2 flux (B) time series, the black line is the 1990-2021 Global Carbon Budget 2022 estimate and the red line is the CMO-NRT near-real-time prediction through July 2023, picking up smoothly where the historical record ends; grey shading spans the range across the ten GOBMs and eight data products. Panels C and D map the mean $f\text{CO}_2$ and flux over August 2022-July 2023.

The red segment in panels A and B is the whole point: it is the part of the curve that, until now, would not have existed for another year. The gridded maps in C and D show that the near-real-time product is not just a global number but a spatially resolved field, with the familiar pattern of CO_2 outgassing in the equatorial Pacific and uptake in the higher-latitude oceans.

Does it faithfully reproduce the originals?

A near-real-time estimate is only useful if it matches what the trusted models would have said. To test this honestly, the authors held out the most recent two years of every model and product, trained only on earlier data, and then predicted the withheld 2020-2021 period so it could be compared against the known originals.

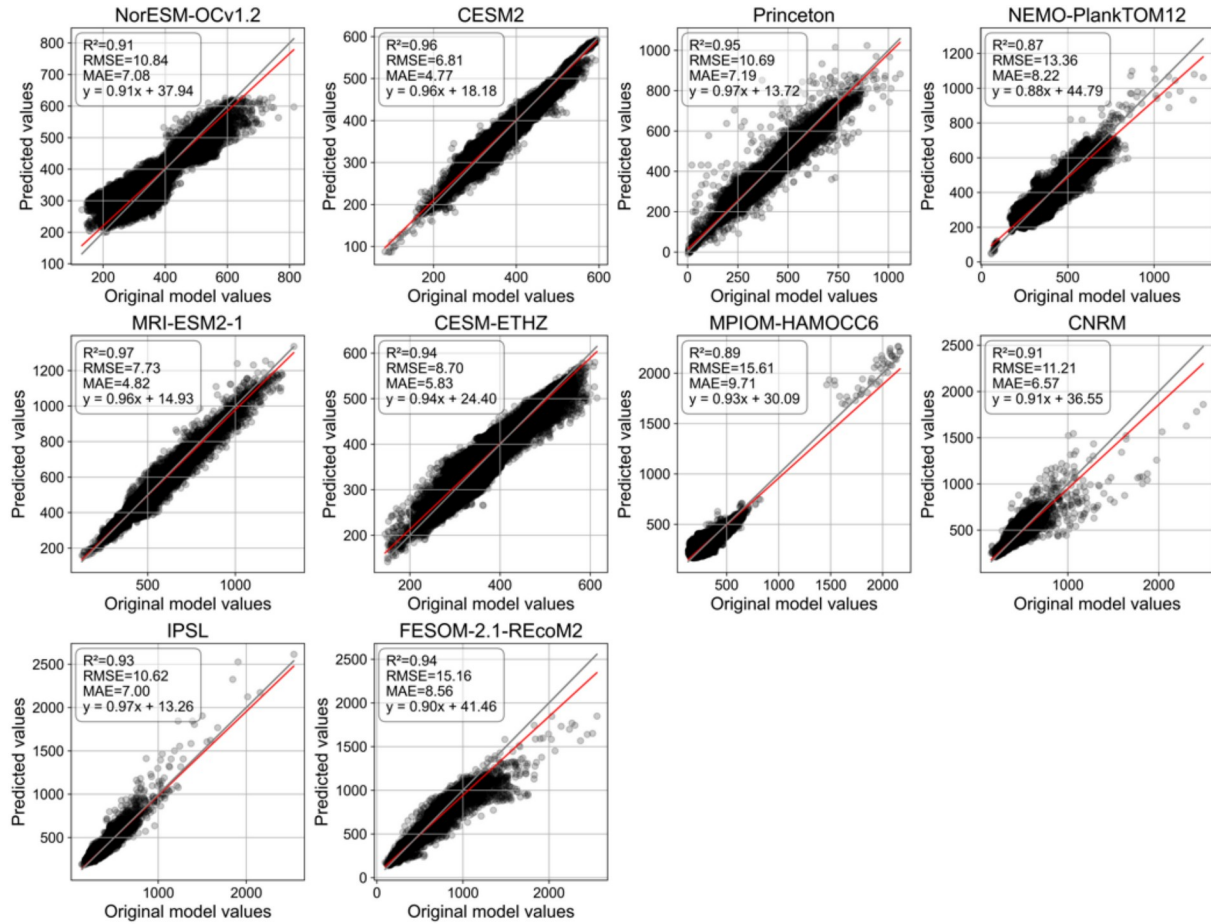


Figure 5. Gridded agreement between CMO-NRT predictions and the original outputs for each of the ten GOBMs over 2020-2021. Points cluster tightly along the 1:1 line (grey), with the total-least-squares fit (red) nearly coincident; per-model correlations run from $R^2 = 0.87$ to 0.97 , most above 0.9 .

For the biogeochemical models the agreement is strong and consistent: the ten GOBMs land between $R^2 = 0.87$ and 0.97 , with several, such as MRI-ESM2-1 (0.97) and CESM2 (0.96), reproduced almost exactly. The table below lists the full set.

Table 1. Gridded correlation (R^2) between CMO-NRT predictions and the original outputs for each of the ten GOBMs over the held-out 2020-2021 period (from Figure 5).

Ocean biogeochemical model (GOBM)	Gridded R^2 (2020-2021)
MRI-ESM2-1	0.97
CESM2	0.96
Princeton	0.95
CESM-ETHZ	0.94
FESOM-2.1-REcoM2	0.94
IPSL	0.93
NorESM-OCv1.2	0.91
CNRM	0.91
MPIOM-HAMOC6	0.89
NEMO-PlankTOM12	0.87

The data products are the harder case, and the paper does not hide it. Compared model-by-model on global monthly averages, some products are reproduced beautifully (NIES-NN and

OSETHZ-GRaCER both reach $R^2 = 0.98$, UOEX-Watson 0.96), while others are noticeably noisier, with occasional points scattering away from the fit line (Jena-MLS at 0.65 and MPI-SOMFFN at 0.53 are the weakest).

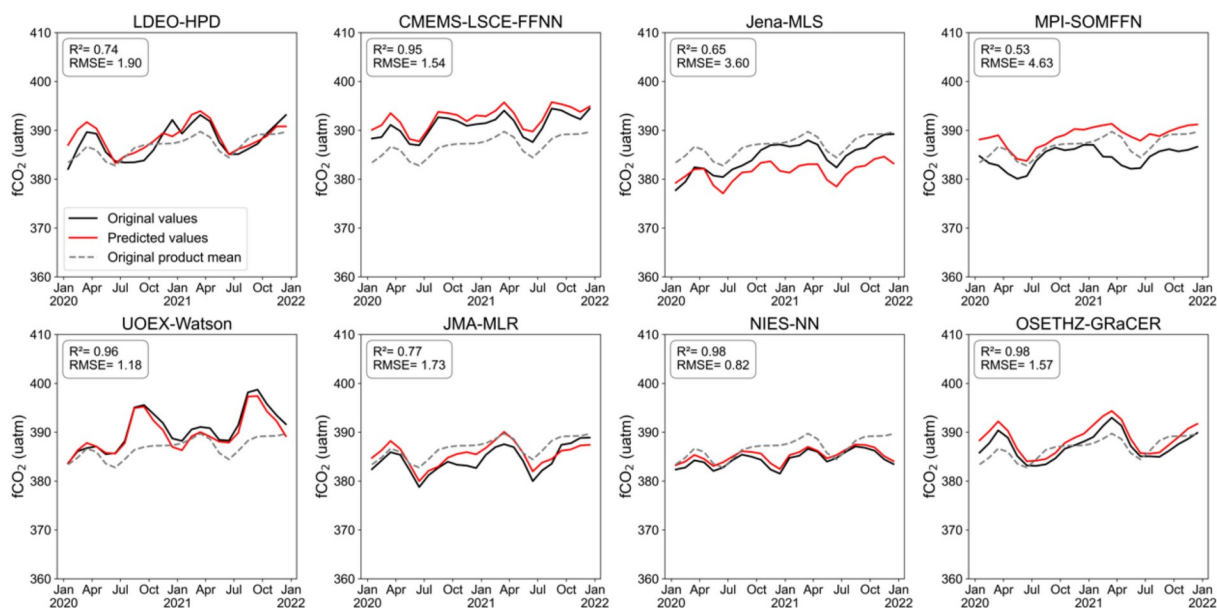


Figure 8. Global-average monthly fCO_2 for each of the eight data products over 2020-2021: original values (black), CMO-NRT predictions (red), and the mean across all products (dashed grey). Agreement is excellent for products like NIES-NN and OSETHZ-GRaCER ($R^2 = 0.98$) but weaker for MPI-SOMFFN ($R^2 = 0.53$), reflecting the larger internal variability of some data products.

Across the global monthly aggregates the predictions run only slightly high, with most differences under 3 uatm, and the aggregate correlations stay mostly above $R^2 = 0.85$. In short, the framework tracks the GOBMs almost perfectly and the stronger data products very well, while remaining candid about the handful of products that are genuinely harder to reproduce.

Where the residuals live

Beyond global averages, it matters whether the small remaining errors are spread evenly or concentrated in a few regions that could bias a basin-scale budget. Mapping the prediction-minus-original differences month by month shows that agreement is broadly uniform across the globe.

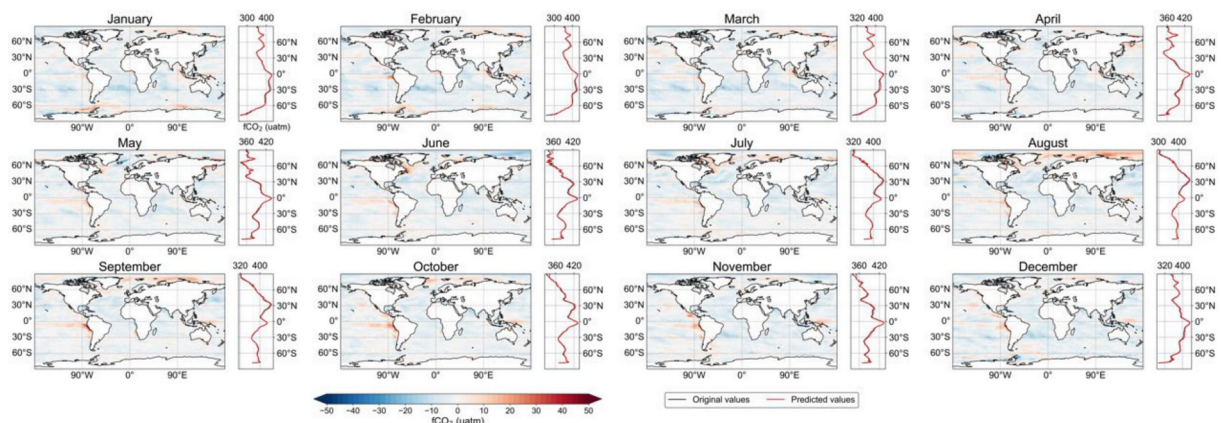


Figure 11. Spatial differences between CMO-NRT predictions and the ten-GOBM average across the twelve months of 2020-2021; side panels show the zonal (latitude) profiles. Residuals are small and largely uniform, with the largest discrepancies concentrated in the Arctic Ocean and the equatorial Pacific.

The maps are mostly pale, meaning small residuals nearly everywhere. What structure remains is concentrated where it is physically hardest to model: the Arctic Ocean, where sea-ice cover and sparse observations complicate the surface CO₂ field, and the equatorial Pacific, where strong upwelling drives large CO₂ gradients. These are exactly the regions to treat with caution when reading the near-real-time product.

Why it matters

CMO-NRT does not reinvent ocean carbon science; it makes the existing, trusted science timely. By pairing convolutional networks with semi-supervised learning to update ten GOBMs and eight data products, it turns a picture of the global ocean carbon sink that used to arrive a year late into a near-real-time, monthly, gridded monitor, here covering January 2022 through July 2023.

For the scientists and policymakers working through the global stocktake, that is the difference between reacting to last year's ocean and watching this year's. The honest boundaries are stated plainly: a couple of noisier data products, and residuals that concentrate in the Arctic and equatorial Pacific. With the dataset and code released openly on Figshare and the project website, the more valuable contribution may be the template itself, a reusable recipe for pulling slow, model-based climate indicators up to the present month.